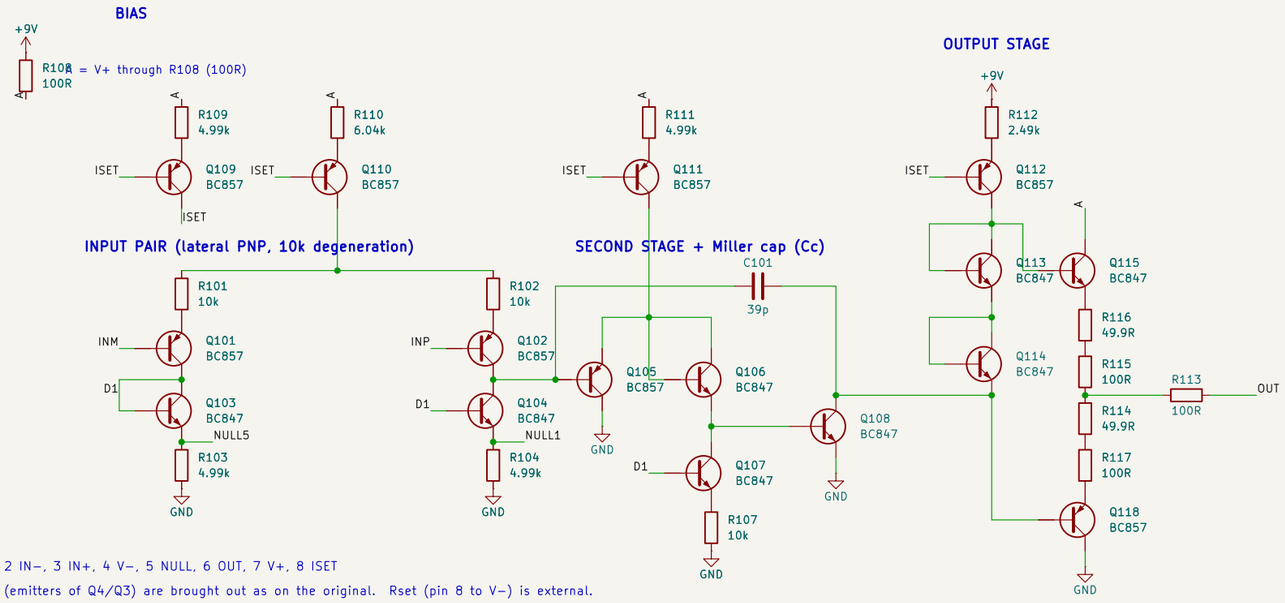
LM4250 CLONE — discrete transistor-level replica of the National/TI LM4250 programmable op-amp (AN-71 fig. 1)

Pin 8 (ISET): resistor to V- sets the bias (Rset = (V+ - 0.5 V) / Iset). Stingray-style preamp: 1M5 at 9 V = 5.7 uA -> GBW ~200 kHz, slew ~0.11 V/us.

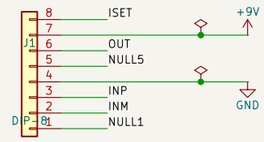
Cc 39 pF (the original 30 pF; 39-40 p tracks the datasheet GBW/slew graphs for Iset 1-20 uA at 9 V and 18 V in simulation) - tune on the bench. Single supply: V- = GND.

Parts are numbered AN-71 number + 100 (Q101..Q118, R101..R117, C101): they never clash with the host pedal's R7, C6 ...



DIP-8 pins: 1 NULL, 2 IN-, 3 IN+, 4 V-, 5 NULL, 6 OUT, 7 V+, 8 ISET

Offset null pins 1/5 (emitters of Q4/Q3) are brought out as on the original. Rset (pin 8 to V-) is external.



From TI AN-71 fig. 1; Rset to V- on pin 8

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Sheet: /

File: LM4250_CLONE.kicad_sch

Title: LM4250 clone (discrete)

Size: A3	Date: 2026-10-25	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 1/1